

Task Scheduler Technical Documentation

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1. Introduction

This document provides technical and operational documentation for the Task Scheduler implemented in the SysTec Smart Thermostat prototype. The Task Scheduler is responsible for managing multiple periodic tasks, ensuring efficient and timely operation of the system's core functionalities.

2. System Overview

The Task Scheduler implements a cooperative multitasking system using a time-driven scheduling approach. It manages three primary tasks: Button Check, Temperature Check, and Server Update.

2.1 Key Components

- Task Structure: Defines properties and state of each task
- Timer: Provides base timing for the scheduler
- Task Array: Holds all tasks to be scheduled

3. Task Structure

```
/* Tasks Structure*/
typedef struct task {
    int state;                // Current state of the task's state machine
    unsigned long period;     // Period of the task in milliseconds
    unsigned long elapsedTime; // Time elapsed since last tick of this task
    int (*TickFct)(int);      // Pointer to the task's tick function
} task;
```

4. Task Scheduler Algorithm

4.1 Initialization

1. Initialize hardware components (GPIO, UART, I2C, Timer)
2. Define tasks with their respective periods and tick functions
3. Initialize task states

4.2 Main Loop

1. For each task in the task array:
 - a. Check if the task's elapsed time has reached its period
 - b. If so, execute the task's tick function and update its state
 - c. Reset the task's elapsed time
2. Increment elapsed time for all tasks by GCD (Greatest Common Divisor) of all periods
3. Wait for next timer interrupt (occurs every GCD milliseconds)

4.3 Inputs

- Timer interrupts (every GCD milliseconds)
- Button presses (via GPIO interrupts)
- Temperature readings (via I2C)

4.4 Outputs

- LED status updates (via GPIO)
- Temperature and system status updates (via UART)

4.5 Expected Results

- Timely execution of all tasks
- Responsive button input handling
- Accurate temperature readings and control
- Regular status updates to the server

5. Task Implementations

5.1 Button Check Task

- Purpose: Monitor button presses to adjust temperature setpoint
- Period: 200ms
- States:
 - BUTTON_INIT: Initialize button state
 - BUTTON_WAIT: Wait for button press
 - BUTTON_INCREASE: Increase setpoint
 - BUTTON_DECREASE: Decrease setpoint

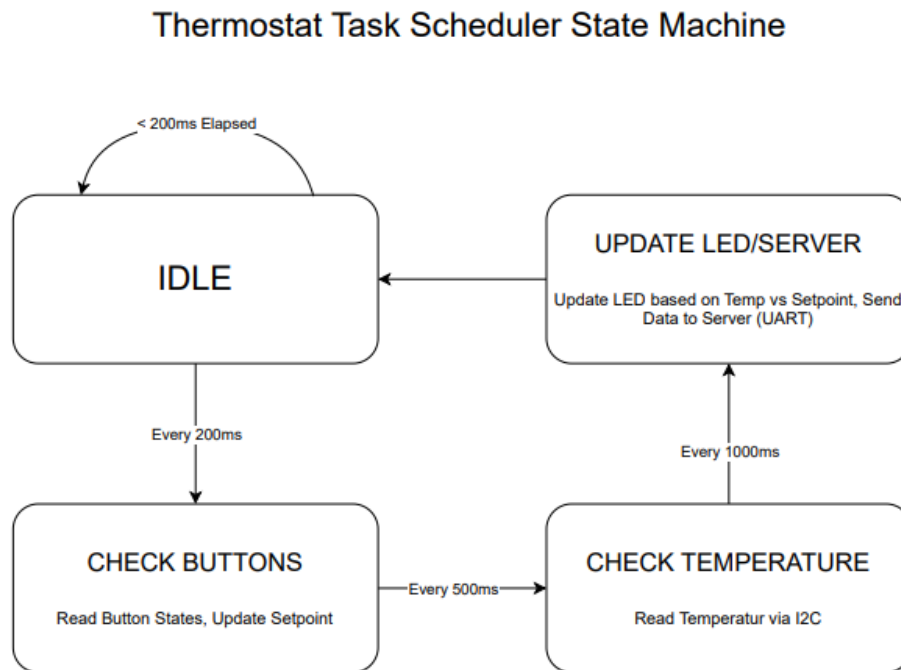
5.2 Temperature Check Task

- Purpose: Read current temperature from sensor
- Period: 500ms
- States:
 - TEMP_INIT: Initialize temperature check
 - TEMP_CHECK: Read current temperature

5.3 Server Update Task

- Purpose: Update server with current system status
- Period: 1000ms
- States:
 - SERVER_INIT: Initialize server update
 - SERVER_UPDATE: Send status update to server

6. State Machine Diagram



7. Conclusion

The Task Scheduler provides an efficient and flexible method for managing multiple periodic tasks in the SysTec Smart Thermostat prototype. Its modular design allows for easy addition of new tasks and modification of existing ones, supporting future expansions of the system's capabilities.

References:

Cooper, Z. (2020, April 6). *A practical guide to writing technical specs*. Stack Overflow.
<https://stackoverflow.blog/2020/04/06/a-practical-guide-to-writing-technical-specs/>